

COMMON PROBLEM DEFINITION

Geometry, effective properties, steady temperature and explicit supports

3D solid elements

27-node solid elements
Full and quarter domains
Uniform / graded meshes

Fourier / 3D reference

Fourier modes in x and y
Independent $U(z)$, $V(z)$, $W(z)$
Quadratic z elements

Sinusoidal plate

Five displacement fields
Sinusoidal shear variation
Plane-stress reduction

VERIFICATION

Patch / thin-plate limits
Thermal stress cross-check

MATCHED RESPONSES

Displacement and stress at common thickness paths
Check Fourier order, thickness and local mesh resolution

RQ1 | Accuracy

Profiles and peaks
Centre / support

RQ2 | Applicability

Thickness / grading
Support distance

RQ3 | Restoration

Positive stiffness
Stress / deflection

RQ4 | Robustness

Properties / pores
Restraints / bounds

Local-model verification; physical calibration and finite-size effects remain separate.